

Mostafa Eslam Elsayed

✉ mostafaeslam1220@gmail.com ☎ [+20 111 214 4903](tel:+201112144903) in [mostafaeslam](#) 🌐 [mostafaeslam](#) 📁 [Portfolio](#)

Professional Summary

AI Engineer with experience in building and deploying machine learning and computer vision systems. Skilled in model development, optimization, and integration using modern frameworks and cloud-based tools. Interested in advancing AI and ML applications through practical, scalable solutions.

Education

Arab Academy for Science, Technology & Maritime Transport

Bachelors in Artificial Intelligence

Alamein, Egypt

Oct 2021 – Jul 2025

- **Major:** Intelligent Systems
- Relevant Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Cloud Computing, Data Structures & Algorithms, HPC.

Experience

Generative AI Professional Trainee

DEPI - Digital Egypt Pioneers Initiative

Alexandria, Egypt

Apr 2024 – Oct 2024

- Completed a 6-month intensive AI program covering **LLMs**, **RAG**, and deployment pipelines.
- Built **3+** generative AI applications using **PyTorch**, **LangChain**, and **Hugging Face**.
- Fine-tuned and deployed models, improving generation quality and inference time by **20%**.

Machine Learning Intern

Mentorless

Remote / Ahmedabad, India

May 2024 – Jun 2024

- Developed disease prediction and market forecasting models using **TensorFlow** and **scikit-learn**.
- Designed and tuned **ML pipelines** for real datasets, achieving **88% accuracy**.
- Delivered a deployable end-to-end workflow demonstrating model readiness for production.

Project Highlights

AI-Based Educational System with Mixed Reality (Graduation Project)

Sep 2024 – Jun 2025

- Built a mixed-reality educational platform with an **AI-powered teaching avatar** using **XREAL Air 2 Ultra**, **Gemini**, **Whisper**, **ElevenLabs**, **LangChain**, and **FastAPI**.
- Integrated **LLM (Gemini)**, **STT**, and **TTS** for interactive, real-time learning.
- Deployed backend with **FastAPI + Docker**, automating slides and quizzes via **Google APIs**.

NLP-Based Code Generation System

Oct 2024

- Developed a text-to-code transformer generating Python/C++ code using **TensorFlow** and **Hugging Face**.
- Fine-tuned CodeT5 on **11K+ samples**, achieving **80% syntactic correctness**.

Face Emotion Recognition Web App

Dec 2023

- Trained **CNN model** on **35K+ FER2013 images** for real-time emotion recognition.
- Deployed via **Flask** and **Docker**, achieving **87% accuracy** and **20 FPS** inference.

Certifications & Training

Generative AI Professional Track — Digital Egypt Pioneers Initiative / MCIT

Apr 2024 – Oct 2024

Machine Learning in Production — DeepLearning.AI / Coursera

Sep 2024

Introduction to Machine Learning — Information Technology Institute / ITI

Aug 2023

Machine Learning Internship — Mentorless

May 2024 – Jun 2024

Technical Skills

Programming & Development: Python, C/C++, Prolog, JavaScript, Node.js, ROS2, MATLAB

Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, LangChain, LangGraph, LangSmith, Streamlit, MCP, FAISS, NLTK

Tools and Methodologies: Git, GitHub, GitLab, Docker, n8n, REST APIs, GCP, AWS, MCP, Prompt Engineering

Core Concepts: MLOps, UML Design, E2E ML Pipelines, Software Documentation, Kinematics